Practice Problem set 11.

Question 1 Let $X, X_1, X_2 \dots$ be real valued r.v. on the same probability space $(\Omega, \mathcal{F}, \mathbb{P})$ Suppose that $X_n \xrightarrow[n\to\infty]{\mathbb{P}} X$. In addition, assume that there exists $Y \in \mathcal{L}^p(\mathbb{P})$ such that $|X_n| \le Y$ a.s. $\forall n$.

Prove that (1) $X \in \mathcal{L}^p(\mathbb{P})$.

(2) $X_n \xrightarrow[n\to\infty]{\mathcal{L}^p(\mathbb{P})} X$.

Proof: (1) $X_n \xrightarrow[n\to\infty]{\mathbb{P}} X$ — given

Then there exists a further subsequence $\{X_{n_k}\}_k$ such that

$$X_n \xrightarrow[n\to\infty]{a.s.} X. \quad \text{—(1)}$$

Also given that: $|X_n| \le Y$. — (2)

$$\Rightarrow |X_{n_k}| \le Y \text{ a.s.} \quad \text{from (1) \& (2).} \quad \text{— (3)}$$

Also given that: $\left(\int |Y|^p \, d\mathbb{P}\right)^{1/p} < \infty$. — (4)

Then, from (3) & (4),

$$\| X_{n_k} - X \|_p \xrightarrow{n\to\infty} 0.$$

$$\Rightarrow \mathbb{E}|X_{n_k}|^p = \mathbb{E}|X|^p < \infty.$$

$$\Rightarrow \left(\mathbb{E}|X|^p\right)^{1/p} < \infty.$$

$$\Rightarrow X \in \mathcal{L}^p(\mathbb{P}).$$

(2) Suppose $X_n \xrightarrow[n\to\infty]{\mathcal{L}^p(\mathbb{P})} X$ does not hold.

Then there exists $\varepsilon > 0$, and a subsequence $\{X_{n_k}\}$ of $\{X_n\}_n$ such that:

$$\mathbb{E}\,|X_{n_k} - X|^p \ge \varepsilon \quad \forall k. \quad \text{— (5)}$$

$$\Rightarrow X_{n_k} \xrightarrow[k\to\infty]{\mathbb{P}} X.$$

Then we have a further subsequence $\{X_{n_{k_\ell}}\}_\ell$ such that $X_{n_{k_\ell}} \xrightarrow[n\to\infty]{a.s.} X$.

But $|X_{n_{k_l}}| \le Y \quad \forall l$ with $Y \in L^p(\mathbb{P})$.

$$\Rightarrow X_{n_{k_l}} \xrightarrow[l \to \infty]{L^p(\mathbb{P})} X$$

$$\therefore \mathbb{E}\,|X_{n_{k_l}} - X|^p \xrightarrow{l \to \infty} 0.$$

This contradicts (5).

Hence proved.

Question 3. Let X be an RV with distribution function F_X. For $c \in \mathbb{R}$ compute $\int_{\mathbb{R}} [F_X(x+c) - F_X(x)]\, d\lambda(x)$

$$\int_{\mathbb{R}} [F_X(x+c) - F_X(x)]\, d\lambda(x)$$

$$= \int_{\mathbb{R}} \left(\mathbb{P}(X \le x+c) - \mathbb{P}(X \le x)\right) d\lambda(x)$$

$$= \int_{\mathbb{R}} \mathbb{P}(x < X \le x+c)\, d\lambda(x)$$

$$= \int_{\mathbb{R}} \left(\int_{\Omega} 1_{(x, x+c]}(X(\omega))\, d\mathbb{P}(\omega) \right) d\lambda(x)$$

$$= \int_{\Omega} \left(\int_{\mathbb{R}} 1_{(x, x+c]}(X(\omega))\, d\lambda(x) \right) d\mathbb{P}(\omega)$$ — by Fubini's theorem.

$$= \int_{\Omega} \lambda((x, x+c])\, d\mathbb{P}(\omega)$$

$$= c \cdot \int_{\Omega} d\mathbb{P}(\omega) = \underline{\underline{c}}.$$

Question 5 Let $X \perp Y$ or defined on $(\Omega, \mathcal{F}, \mathbb{P})$.

Let $X, Y \in L^2(\mathbb{P})$ $\quad Var(X+Y) = Var(X) + Var(Y)$

$\mathbb{E}(X+Y - \mathbb{E}(X+Y))^2$.

$= \mathbb{E}((X+Y)^2) - (\mathbb{E}(X+Y))^2$

$= \mathbb{E}(X^2 + Y^2 + 2XY) - \{(\mathbb{E}X)^2 + (\mathbb{E}Y)^2 + 2(\mathbb{E}X)(\mathbb{E}Y)\}$.

$= \mathbb{E}X^2 + \mathbb{E}Y^2 + 2\mathbb{E}XY - (\mathbb{E}X)^2 - (\mathbb{E}Y)^2 - 2(\mathbb{E}X)(\mathbb{E}Y)$

As $X \perp Y \Rightarrow \mathbb{E}XY = \mathbb{E}X\,\mathbb{E}Y$

$= \mathbb{E}X^2 - (\mathbb{E}X)^2 + \mathbb{E}Y^2 - (\mathbb{E}Y)^2$

$= Var(X) + Var(Y)$ QED.

Question 4. Find examples of RVs X and Y, defined on the same probability space which are uncorrelated but not independent.

$\rightarrow X = \{-1, 0, 1\}$.

$\mathbb{P}(X=-1) = \mathbb{P}(X=0) = \mathbb{P}(X=1) = 1/3$.

Let $Y = X^2$.

$\therefore \mathbb{P}(Y=1) = 2/3 \quad \mathbb{P}(Y=0) = 1/3$.

$\mathbb{E}X = -\frac{1}{3} + \frac{1}{3} + 0 = 0$

$\mathbb{E}Y = \frac{2}{3}$

$\mathbb{E}XY = \mathbb{E}X \cdot X^2 = \mathbb{E}X^3 = -\frac{1}{3} + \frac{1}{3} = 0$.

$\therefore \mathbb{E}XY = \mathbb{E}X\,\mathbb{E}Y = 0$.

but $Y = X^2$ hence they are not independent

Question 1 X and Y are independent iff $\sigma(X)$ & $\sigma(Y)$

$\sigma(X) = \{X^{-1}(A) \mid A \in \mathcal{B}_{\mathbb{R}}\}$,

$\sigma(Y) = \{Y^{-1}(A) \mid A \in \mathcal{B}_{\mathbb{R}}\}$.

$\sigma(X)$ & $\sigma(Y)$ are indepent.

$\Leftrightarrow \mathbb{P}(X^{-1}(A) \cap Y^{-1}(B)) = \mathbb{P}(X^{-1}(A))\mathbb{P}(Y^{-1}(B)) \quad \forall A, B \in \mathcal{B}_{\mathbb{R}}$

$\Leftrightarrow \mathbb{P}(X \in A, Y \in B) = \mathbb{P}(X \in A)\mathbb{P}(Y \in B)$, $\forall A, B \in \mathcal{B}_{\mathbb{R}}$.

$\Leftrightarrow$ X & Y are independent

(b) X & Y are indepoendent. f, g are measurable.

Then independence of X & Y implies the independence of $f(X)$ & $f(Y)$.

$\Rightarrow \mathbb{P}(X \in A, Y \in B) = \mathbb{P}(X \in A)\mathbb{P}(Y \in B) \quad \forall A, B \in \mathcal{B}_{\mathbb{R}}$.

Now, $\mathbb{P}(f(X) \in A, g(Y) \in B) = \mathbb{P}(X \in f^{-1}(A), Y \in g^{-1}(B))$

As f, g are measurable.

$\forall A, B \in \mathcal{B}_{\mathbb{R}}$, $f^{-1}(A), g^{-1}(B) \in \mathcal{B}_{\mathbb{R}}$.

$\therefore \mathbb{P}(X \in f^{-1}(A), Y \in g^{-1}(B)) \quad \forall A, B \in \mathcal{B}_{\mathbb{R}}$

$= \mathbb{P}(X \in f^{-1}(A), Y \in g^{-1}(B)) \quad \forall f^{-1}(A), g^{-1}(B) \in \mathcal{B}_{\mathbb{R}}$.

$= \mathbb{P}(X \in f^{-1}(A))\mathbb{P}(Y \in g^{-1}(B)) \quad \forall f^{-1}(A), g^{-1}(B) \in \mathcal{B}_{\mathbb{R}}$.

$= \mathbb{P}(f(X) \in A) \cdot \mathbb{P}(g(Y) \in B)$, $\forall A, B \in \mathcal{B}_{\mathbb{R}}$.

$\therefore f(X)$ & $g(Y)$ are indepenedent if X & Y are indepenedent.

Question 2 Given an RV X and $\alpha \geq 1$, then:

(a) $\mathbb{E}|X|^{\alpha} = \int_0^{\infty} \mathbb{P}(|X|^{\alpha} > x)\, dx = \alpha \int_0^{\infty} x^{\alpha-1} \mathbb{P}(|X| > x)\, dx$.

(a) $\mathbb{E}|X|^{\alpha} = \int_0^{\infty} \mathbb{P}(|X|^{\alpha} > x)\, dx$

Let $Y = |X|^{\alpha}$.

$\mathbb{E}Y = \int_{\Omega} |X|^{\alpha}(\omega)\, d\mathbb{P}(\omega) = \int_{\Omega} \left(\int_{\mathbb{R}} 1_{[0, x(\omega))}(x)\, d\lambda(x) \right) d\mathbb{P}(\omega)$.

by Fubini's thm,

$= \int_{(0,\infty)} \left(\int_{\Omega} 1_{[0, |X(\omega)|)}(x)\, d\mathbb{P}(\omega) \right) d\lambda(x)$.

$$= \int_{(0,\infty)} \mathbb{P}(|X|^\alpha > x)\, d\lambda(x)$$

$$= \int_0^\infty \mathbb{P}(|X|^\alpha > x)\, dx$$

(b) To show: $\int_0^\infty \mathbb{P}(|X|^\alpha > x)\, dx = \alpha \int_0^\infty x^{\alpha-1} \mathbb{P}(|X| > x)\, dx$.

$$\alpha \int_0^\infty x^{\alpha-1} \mathbb{P}(|X| > x)\, dx$$

$$= \int_0^\infty \alpha x^{\alpha-1} \mathbb{P}(|X| > x)\, dx.$$

$$= \int_0^\infty \alpha\, x^{\alpha-1} \mathbb{P}(|X|^\alpha > x^\alpha)\, dx. \quad [\text{inequality remains same as function is } \uparrow]$$

$$= \int_0^\infty \alpha x^{\alpha-1} \mathbb{P}(|X|^\alpha > x^\alpha)\, dx$$

Now let $x^\alpha = u$.

$\Rightarrow \alpha x^{\alpha-1} dx = du$

$$= \int_0^\infty \mathbb{P}(|X|^\alpha > u)\, du.$$

Let $u = x$.
$du = dx$.

$$= \int_0^\infty \mathbb{P}(|X|^\alpha > x)\, dx.$$

(ii) To show: $i.f. \; \mathbb{E}|X|^\alpha < \infty \iff \sum_{n=0}^\infty \mathbb{P}(|X|^\alpha > n) < \infty.$

$$\iff \sum_{n=0}^\infty \mathbb{P}(|X| > n^{1/\alpha}) < \infty.$$

Proof: (a) $\sum_{n=0}^\infty \mathbb{P}(|X|^\alpha > n) < \infty. \iff \sum_{n=0}^\infty \mathbb{P}\left((|X|^\alpha)^{1/\alpha} > n^{1/\alpha}\right) < \infty.$

$\iff \sum_{n=0}^\infty \mathbb{P}(|X| > n^{1/\alpha}) < \infty.$

(b)

$\mathbb{E}|X|^\alpha < \infty \iff \sum_{n=1}^{\infty} \mathbb{P}(|X|^\alpha > n) < \infty.$

$$\mathbb{E}|X|^\alpha = \int_0^\infty \mathbb{P}(|X|^\alpha > x)\, dx.$$

$$= \int_{[0,\infty)} \mathbb{P}(|X|^\alpha > x)\, dx.$$

$$= \int_{\mathbb{R}} \mathbb{P}(|X|^\alpha > x)\, 1_{[0,\infty)}(x)\, d\lambda(x)$$

$$1_{[0,\infty)}(x) = \sum_{n=0}^{\infty} 1_{[n,n+1)}(x)$$

$$= \int_{\mathbb{R}} \mathbb{P}(|X|^\alpha > x) \sum_{n=0}^{\infty} 1_{[n,n+1)}\, d\lambda(x)$$

$$= \int_{\mathbb{R}} \sum_{n=0}^{\infty} \mathbb{P}(|X|^\alpha > x) \cdot 1_{[n,n+1)}\, d\lambda(x). \quad \text{by MCT.}$$

$$= \sum_{n=0}^{\infty} \int_{[n,n+1)} \mathbb{P}(|X|^\alpha > x)\, d\lambda(x)$$

For $x \in [n, n+1)$

$$\Rightarrow \mathbb{P}(|X|^\alpha > n) \geq \mathbb{P}(|X|^\alpha > x) \geq \mathbb{P}(|X|^\alpha > n+1)$$

$$\Rightarrow \sum_{n=0}^{\infty} \mathbb{P}(|X|^\alpha > n) \geq \sum_{n=0}^{\infty} \mathbb{P}(|X|^\alpha > x) \geq \sum_{n=0}^{\infty} \mathbb{P}(|X|^\alpha > n+1)$$

$$\Rightarrow \sum_{n=0}^{\infty} \mathbb{P}(|X|^\alpha > n) \geq \mathbb{E}|X|^\alpha \geq \sum_{n=0}^{\infty} \mathbb{P}(|X|^\alpha > n+1)$$

$\therefore \mathbb{E}|X|^\alpha < \infty \Rightarrow \sum_{n=0}^{\infty} \mathbb{P}(|X|^\alpha > n) < \infty.$

$$\lim_{n \to \infty} \mathbb{P}(|X|^\alpha > n) = 0.$$

(c) $\lim_{n\to\infty} n^{\alpha} P(|X|>n) = 0$.

$\lim_{n\to\infty} n^{\alpha} P(|X|^{\alpha} > n^{\alpha}) = 0$. by Abel's (Pringsheim) theorem

$\Rightarrow \lim_{n\to\infty} n^{\alpha} P(|X|^{\alpha} > n) = 0$.

Abel's (Pringsheim) theorem: (1) ~~If~~ terms of series.
(2) Convergent
(3) Monotonically decreasing
$\sum_{n=1}^{\infty} a_n$ then $\lim_{n\to\infty} n a_n = 0$.